

Q3		
(a)	A polarised wave is one in which the <u>vibrations/oscillations of the wave are restricted to only one direction</u> <u>in the plane normal/perpendicular to the direction of energy transfer.</u>	B1 B1
(b)(i)	As long as polarising filter B is perpendicular to either polarising filter A or filter C, the emergent light from filter C will be zero. Hence, the other angle that occurs will be when polarising axis of B is 90° of C, i.e. <u>$\theta = 135^\circ$</u>	A1
(b)(ii)	Using Malus' Law, Intensity of light emergent from polarising filter B = $I \cos^2(60^\circ)$ Intensity of light emergent from polarising filter C = $I \cos^2(60^\circ) \cos^2(60^\circ - 45^\circ)$ $= 0.233 I$	M1 A1

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